Foundation Models for Satellite Earth Observation

Abstract

The exponential proliferation of Earth Observation (EO) data, driven by rapid advancements in satellite sensor technology and large-scale orbital deployments, has fundamentally transformed geospatial analysis. Historically, remote sensing applications relied on task-specific supervised deep learning models, which were severely constrained by the scarcity of high-quality, pixel-level annotations and the inherent heterogeneity of multi-sensor data. The advent of foundation models—large-scale neural architectures pre-trained on vast, unannotated datasets using self-supervised learning—has precipitated a paradigm shift in geospatial artificial intelligence. This comprehensive survey investigates the evolution of foundation models in satellite Earth observation, tracing the trajectory from unimodal optical image encoders to highly complex, multimodal, and spatiotemporal systems.
This manuscript systematically reviews the architectural foundations underlying these models, contrasting the dominant Vision Transformer (ViT) paradigms with emerging linear-complexity State Space Models (SSMs), such as Mamba, which are uniquely suited for the high-resolution demands of geospatial data. Furthermore, the survey explores pre-training methodologies, including Masked Image Modeling (MIM) and Contrastive Learning, and details their specialized adaptations for multispectral, hyperspectral, and Synthetic Aperture Radar (SAR) modalities. The integration of vision-language and vision-location models is analyzed, highlighting their capacity for zero-shot retrieval, open-vocabulary reasoning, and implicit spatial representation. Finally, this survey critically examines the current landscape of geospatial benchmarking, addressing the urgent need for standardized evaluation protocols, robustness testing under distributional shifts, and transparent model weight release to resolve the systemic ambiguities currently obscuring the true state of the art in geospatial foundation models.

Introduction

Recent advancements in Earth Observation (EO) technologies have revolutionized the capacity of the scientific community to monitor planetary-scale dynamics. Modern satellite constellations, ranging from the Copernicus Sentinel missions to Landsat and specialized hyperspectral instruments like the Environmental Mapping and Analysis Program (EnMAP) and PRISMA, generate terabytes of high-dimensional data daily[1]. This influx of data provides critical spatial, temporal, and spectral information essential for addressing global challenges, including land-use and land-cover (LULC) classification, climate change mitigation, disaster response, precision agriculture, and the broader United Nations Sustainable Development Goals (SDGs)[4]. However, the sheer volume and complexity of this data—encompassing multi-spectral bands extending into the shortwave infrared (SWIR), polarimetric synthetic aperture radar (SAR) backscatter, and high-frequency spatiotemporal time series—render traditional analytical methods inadequate[7].

Traditionally, the remote sensing community relied on supervised deep learning models, particularly Convolutional Neural Networks (CNNs) and early Transformer architectures, engineered for narrow, specific tasks[4]. This paradigm required massive volumes of meticulously annotated data. Because EO data analysis often necessitates pixel-level semantic segmentation across vast geographic areas, data annotation became an exorbitant bottleneck requiring extensive domain expertise[11]. Consequently, the deep learning models trained under this supervised paradigm suffered from task fragmentation. They functioned as isolated computational silos, incapable of generalizing across diverse sensors, varying geographic topographies, or shifting environmental conditions, thereby limiting their operational utility[14].
The introduction of Foundation Models (FMs) offers a definitive solution to this scalability and generalization crisis. Originally popularized in natural language processing through Large Language Models (LLMs), foundation models are defined as highly parameterized neural networks pre-trained on broad, uncurated datasets primarily through self-supervised learning (SSL)[17]. The fundamental objective of SSL is to extract generic, task-agnostic representations from unlabeled data. In the context of remote sensing, Geospatial Foundation Models (GFMs) exploit the vast, openly available archives of satellite imagery to capture complex spatial, spectral, and temporal dependencies[18]. Once pre-trained, these foundational representations can be efficiently adapted to a multitude of downstream tasks via parameter-efficient fine-tuning (PEFT), linear probing, or zero-shot prompting, establishing a highly efficient "pre-train once, adapt many" methodology[18].
Despite rapid conceptual progress and promising benchmark results, the application of foundation models in remote sensing remains relatively nascent and highly fragmented[7]. Significant technical challenges persist regarding cross-modal alignment between radically different imaging physics (e.g., optical reflectance versus SAR microwave backscatter), the computational intractability of processing massive hyperspectral cubes with quadratic-complexity attention mechanisms, and the pervasive lack of standardized, rigorous evaluation frameworks[23]. This academic survey provides a systematic, highly detailed analysis of the current state of the art in Geospatial Foundation Models. The subsequent sections organize the research landscape by architectural innovation, pre-training objectives, modality fusion strategies, vision-language integration, and the critical evaluation of model robustness and deployment prospects.

Methods

The landscape of Geospatial Foundation Models is exceptionally diverse, necessitating a structured taxonomic approach to understand the underlying mechanisms driving recent innovations. This section organizes the research area into coherent method families, beginning with the foundational neural architectures, progressing through self-supervised pre-training paradigms, and culminating in specialized models tailored for distinct sensor modalities and reasoning tasks.

Architectural Foundations: Transformers and State Space Models

The core architectures utilized in remote sensing have largely mirrored broader developments in the computer vision domain. However, significant modifications are required to accommodate the high dimensionality, varied spatial resolutions, and non-causal nature of Earth observation data.

Vision Transformers and the Attention Bottleneck

The Vision Transformer (ViT) architecture fundamentally disrupted geospatial representation learning by abandoning the inductive biases inherent in CNNs—such as translation invariance and strictly local receptive fields—in favor of global self-attention mechanisms[17]. In a standard ViT-based GFM, a multi-channel satellite image is partitioned into a sequence of non-overlapping patches. These patches are linearly projected into a latent token space, summed with positional embeddings that encode spatial coordinates, and processed through alternating layers of Multi-Head Self-Attention (MHSA) and Multi-Layer Perceptrons (MLPs)[17]. This global receptive field allows ViTs to model long-range dependencies, such as the spatial relationship between a distant river and an agricultural field, which is critical for accurate scene understanding. However, the standard ViT architecture exhibits a critical mathematical limitation: the computational complexity and memory footprint of the self-attention mechanism scale quadratically with the sequence length, denoted as $\mathcal{O}(N^2)$ [27]. Because remote sensing tasks frequently require processing exceptionally high-resolution imagery (often exceeding $10,000 \times 10,000$ pixels) or long temporal sequences to detect fine-grained structural features, the sequence length N becomes prohibitively large, creating a severe computational bottleneck that restricts the scalability of purely attention-based models[29].

State Space Models (SSMs) and Vision Mamba

To resolve the quadratic complexity limitation of Transformers while preserving global context modeling, the remote sensing community has rapidly adopted structured State Space Models (SSMs), specifically the Mamba architecture[30]. Mamba provides linear-time sequence processing, $\mathcal{O}(N)$, effectively solving the memory constraints associated with high-resolution satellite imagery[30].

Mamba is theoretically rooted in continuous-time state space formulations, where a one-dimensional input sequence $x(t)$ is mapped to an output sequence $y(t)$ via a hidden latent state $h(t)$, governed by continuous parameters. To adapt this continuous system to discrete image tokens, the formulation is discretized using a Zero-Order Hold (ZOH) operation[31]. The breakthrough innovation of Mamba is the Selective State Space mechanism (S6). Unlike prior SSMs with static transition matrices, S6 parameterizes the state matrices based directly on the input data. This input-dependent selective scan enables the model to dynamically remember task-relevant spatial information (e.g., building footprints) and forget irrelevant background noise (e.g., uniform cloud cover) indefinitely across the sequence[30].

In geospatial applications, adapting Mamba requires overcoming the unidirectional nature of its recurrent scan, which was originally designed for autoregressive language modeling. Remote sensing images are non-causal and exhibit omnidirectional spatial dependencies. Consequently, models such as RSMamba and SatMamba introduce dynamic multi-path and bidirectional scanning mechanisms to process two-dimensional spatial features comprehensively, ensuring that context is aggregated from all directions[10].

Furthermore, advanced frameworks like RoMA (Rotation-aware Multi-scale Autoregressive learning) customize the Mamba architecture specifically for remote sensing pre-training. Because geospatial objects—such as ships, airplanes, or agricultural pivots—appear in arbitrary orientations from an overhead perspective without a fixed "up" vector, standard sequential scanning introduces orientation bias. RoMA incorporates adaptive region cropping and angle-aware embeddings, forcing the state space model to predict angular variations during autoregressive pre-training. This mechanism embeds rotation invariance directly into the Mamba backbone, yielding highly robust representations for oriented object detection[34]. Similarly, the SAMBA framework (Scattering-guided bidirectional Mamba) adapts SSMs for SAR target interpretation, utilizing a three-level hierarchical masking strategy guided by SAR physical priors to mitigate the disruptive effects of speckle noise while maintaining linear complexity[27].

Self-Supervised Pre-training Paradigms

Given the high cost of manual annotation, Self-Supervised Learning (SSL) serves as the fundamental engine for training GFMs. The two dominant paradigms in the remote sensing literature are Contrastive Learning and Masked Image Modeling (MIM).

Contrastive Learning Frameworks

Contrastive learning architectures—including frameworks such as SimCLR, MoCo, BYOL, and SwAV—learn robust representations by minimizing the latent distance between differently augmented views of the same image (positive pairs) while maximizing the distance from all other images in a batch (negative pairs)[11].

In the remote sensing domain, temporal and multi-sensor data provide naturally occurring augmentations that far exceed standard computer vision techniques like color jittering or cropping. For instance, the SeCo (Seasonal Contrast) model utilizes multi-temporal images of the identical geographic location acquired during different seasons as positive pairs. By pulling these representations together, the network is forced to learn invariant semantic features (e.g., urban infrastructure, permanent water bodies) while disregarding transient seasonal phenological changes or temporary snow cover[12]. Similarly, cross-modal contrastive models such as CROMA utilize paired optical and SAR imagery of the exact same geographical footprint. By optimizing a contrastive objective between the optical and radar views, the model maps the disparate physical measurements—solar reflectance and microwave backscatter—into a shared, highly correlated latent space[37].

Masked Image Modeling (MIM)

While contrastive learning excels at establishing global, image-level representations, Masked Image Modeling (MIM) has proven vastly superior for dense, pixel-level prediction tasks such as semantic segmentation and change detection. MIM forces the model to learn fine-grained, localized spatial relationships[21].

The Masked Autoencoder (MAE) framework is highly adaptable to the unique dimensionalities of Earth observation data. In this approach, a substantial fraction of the input image patches (typically ranging from 50% to 90%) are randomly masked and withheld from the encoder. A lightweight decoder is then tasked with reconstructing the original pixel values or frequency domains of the masked regions based solely on the latent representations of the visible patches and learnable mask tokens[28]. This generative pre-training task forces the network to deeply understand spatial continuity and object structure. Notable adaptations of the MAE framework include:

- **Scale-Aware Masking:** FMs designed for multi-sensor data must address the drastically varying Ground Sample Distances (GSD) inherent in satellite collections (e.g., 10m Sentinel-2 versus 0.3m Maxar imagery). The Scale-MAE model resolves this by utilizing the actual geographic area covered by an image (in square meters) to determine the scale of the Vision Transformer's positional encodings, rather than relying on raw pixel resolution. During pre-training, Scale-MAE decodes the masked image through a Laplacian bandpass filter to reconstruct explicit high-frequency and low-frequency images at different conceptual scales, yielding highly robust multi-scale geospatial representations[41].
- **Spatiotemporal Masking:** To process time-series data, advanced MAE models construct three-dimensional "tubelets" comprising height, width, and time dimensions. Masking is applied across both spatial and temporal axes, forcing the neural network to interpolate missing phenological data or cloud-occluded pixels based on the surrounding spatial context and the historical temporal trajectory[40].

Multispectral and Multi-Temporal Foundation Models

Multispectral imagery (MSI) acquired from prominent public missions, such as the Sentinel-2 constellation and the Landsat series, forms the foundational backbone of modern Earth observation. Foundation models designed for MSI must possess the capability to process continuous spectral bands extending into the Near-Infrared (NIR) and Shortwave Infrared (SWIR), capturing data that is entirely invisible to standard RGB models.

The Prithvi-EO Framework

The Prithvi-EO family of models, developed through a collaboration between NASA and IBM Research, represents a highly mature implementation of multi-temporal GFMs. The initial Prithvi-EO-1.0 and its subsequent iteration, Prithvi-EO-2.0 (scaling up to 600 million parameters), are pre-trained on over 4.2 million global time-series samples extracted from the Harmonized Landsat and Sentinel-2 (HLS) archive at a consistent 30-meter spatial resolution[45].

Prithvi-EO employs a highly advanced spatiotemporal Masked Autoencoder architecture. To optimize representation learning, Prithvi-EO-2.0 introduced a multi-objective continual pre-training paradigm that utilizes a teacher-student knowledge distillation structure. A frozen teacher model, originally pre-trained on massive natural image datasets (e.g., ImageNet-22k), provides rich intermediate feature targets. Concurrently, the randomly initialized student model is jointly trained to minimize a traditional pixel-wise MAE reconstruction loss ($\mathcal{L}_{MIM}$) and a feature distillation cosine similarity loss ($\mathcal{L}_{feat}$)[40]. This hybrid pre-training objective allows the geospatial model to inherit the robust structural and edge-detection priors of established computer vision models while adapting strictly to the distinct radiometric and spectral properties of satellite data. Furthermore, Prithvi-EO-2.0 incorporates dynamic temporal and geolocation embeddings. By fusing center latitude, longitude, and acquisition date metadata into the token embeddings using 2D sine/cosine functions, the model achieves significantly enhanced performance on geographically localized tasks, such as regional crop-type classification[40].

SatMAE and Presto

The SatMAE architecture extends the fundamental MAE framework to multispectral and multi-temporal data by grouping subsets of spectral bands (e.g., grouping Visible, NIR, and SWIR channels) and applying independent, learnable spectral positional encodings to each group[49]. This architectural decision prevents the transformer from conflating highly correlated adjacent bands and allows the model to flexibly process varying channel configurations depending on the downstream application.

Conversely, the Presto (Pretrained Remote Sensing Transformer) model is designed specifically for highly efficient, pixel-level time-series analysis[52]. Rather than treating geospatial data as traditional 2D spatial image patches, Presto operates exclusively on individual 1D pixel-timeseries, learning solely from the temporal trajectory of spectral reflectance. Presto reconstructs structurally masked temporal sequences during pre-training, making it exceptionally resilient to pervasive EO challenges such as severe cloud occlusion or missing sensor data. Presto operates with up to 1000$\times$ fewer trainable parameters than massive spatial ViTs, yet achieves state-of-the-art results in phenological downstream tasks, such as agriculture and forest monitoring, facilitating scalable deployment on resource-constrained hardware[16]. A similar paradigm is observed in EarthPT, a 700 million parameter decoding transformer foundation model trained in an autoregressive self-supervised manner specifically for forecasting future pixel-level surface reflectances[54].

Hyperspectral Foundation Models

Hyperspectral Imaging (HSI) instruments—such as EnMAP, PRISMA, and airborne AVIRIS sensors—capture hundreds of extremely narrow, contiguous spectral bands[2]. This hyper-dimensionality provides immense physical detail regarding material composition, biochemical properties, and subtle atmospheric variables[3]. However, the extreme data volume and spectral redundancy of HSI pose severe computational and architectural challenges for standard foundation models.

The SpectralGPT model addresses this dimensionality by employing a novel 3D generative pre-trained transformer architecture. It introduces 3D tokenization to explicitly model spatial-spectral coupling, enabling the transformer to process hyperspectral cubes of varying spatial sizes, spatial resolutions, and arbitrary band counts in a progressive training fashion[28].

To alleviate the historical scarcity of HSI pre-training data, the SpectralEarth dataset and its associated foundation models compiled nearly 3.3 terabytes of globally distributed, cloud-free EnMAP observations, comprising over 538,000 hyperspectral image patches[2]. Using this massive corpus, researchers successfully integrated lightweight convolutional spectral adapters into classical vision backbones, allowing standard ViT and CNN architectures to accommodate the unique characteristics of HSI data[2].

To further mitigate the immense computational load of processing hundreds of channels, parameter-efficient architectures like HyperFM have been developed. HyperFM introduces specialized Hypoformer blocks equipped with intra-group and inter-group spectral attention mechanisms. By employing hybrid parameter decomposition, HyperFM captures complex spectral-spatial relationships using approximately 50% fewer parameters than baseline HSI models, achieving superior performance on atmospheric cloud-property retrieval tasks[3]. Similarly, the HyperFree framework presents a revolutionary "tuning-free" paradigm. It utilizes a vast spectral weight dictionary that dynamically generates embedding layers based strictly on the specific input wavelengths of an unseen HSI sensor. This allows the model to process diverse HSI inputs directly via visual prompt engineering, entirely circumventing the need for computationally ruinous, image-by-image fine-tuning[8].

Multi-Sensor and Any-to-Any Foundation Models

Because operational Earth observation applications frequently demand the synthesis of data from multiple, disparate sensors—such as combining optical multispectral imagery for biochemical vegetation mapping with cloud-penetrating SAR for continuous structural monitoring—the latest generation of GFMs focuses on universal, multi-sensor integration.

Model Name	**Primary Modalities**	**Core Architectural Innovation**	**Key Capabilities**
Prithvi-EO-2.0	HLS (Optical/MSI)	Multi-objective continual distillation; 3D spatiotemporal ViT	High temporal robustness; geolocation embeddings
SpectralGPT	Hyperspectral	3D tokenization for spatial-spectral coupling	Progressive training on varying band counts
Copernicus-FM	Optical, SAR, Met.	Dynamic hypernetworks; metadata integration	Unifies surface and atmospheric observations

TerraMind	9 Modalities (Any)	Finite Scalar Quantization (FSQ); dual-scale encoding	"Thinking in Modalities" (TiM) missing data generation
Clay v1.5	Optical, SAR, DEM	Wavelength-aware continuous embedding mappings	Sensor-agnostic dynamic embedding blocks

DOFA and Copernicus-FM

The DOFA (Dynamic One-For-All) architecture introduced a paradigm shift by demonstrating that a single, unified backbone can effectively process imagery from entirely different sensors (Sentinel-1, Sentinel-2, Gaofen, Landsat) without requiring separate, modality-specific encoders[38].
Building significantly upon this foundation, Copernicus-FM utilizes extended dynamic hypernetworks. Pre-trained on the massive *Copernicus-Pretrain* dataset—which integrates 18.7 million aligned images across all major Sentinel 1-5P missions—Copernicus-FM dynamically adapts its patch embeddings to handle arbitrary spectral or non-spectral modalities (such as atmospheric aerosol concentrations or SAR backscatter). This architectural flexibility breaks the historical, siloed approach of treating Earth surface observation and atmospheric monitoring as fundamentally separate domains, enabling holistic Earth system modeling[25].

TerraMind and "Thinking in Modalities"

TerraMind, developed collaboratively by IBM Research, ESA Φ-lab, and the Jülich Supercomputing Centre, represents the first true "any-to-any" generative, multimodal foundation model for Earth observation[62]. TerraMind is pre-trained on a highly structured dual-scale representation, simultaneously processing token-level and pixel-level data across nine distinct modalities, including Sentinel-1, Sentinel-2, Digital Elevation Models (DEM), and categorical Land Use Land Cover (LULC) maps[63].
The most profound innovation introduced by TerraMind is an inferential capability termed **Thinking in Modalities (TiM)**. When fine-tuned for a specific downstream task with incomplete sensor data (e.g., performing precise flood boundary segmentation using only SAR backscatter data), the model can temporarily pause its standard forward pass. It utilizes its learned cross-modal correlations to internally *generate* an artificial intermediate modality (e.g., a synthesized LULC map or NDVI vegetation index) directly within its token space. It then appends these "imagined" tokens to its own input sequence before executing the final prediction[62]. Because the generated modalities exist purely as latent tokens discretized via Finite-Scalar-Quantization (FSQ), TiM entirely avoids the computational overhead of decoding the prediction back into pixel space. This allows TerraMind to significantly boost downstream

accuracy—such as improving flood segmentation mIoU by over 2 percentage points—without requiring the physical presence of the secondary modality at inference time[62].

Vision-Language Models (VLMs) and Location Encoders

The inability of purely visual foundation models to interface with human instructions or process natural language queries has spurred intense development in Remote Sensing Vision-Language Models (RSVLMs). Standard models like OpenAI's CLIP cannot be directly applied to EO data due to a severe domain gap and the acute lack of massive, high-quality paired image-text datasets specific to satellite imagery[68].

RSVLMs and Dual-Granularity Alignment

To mitigate the data scarcity bottleneck, RemoteCLIP employs a programmatic Box-to-Caption (B2C) and Mask-to-Box (M2B) conversion strategy. This approach systematically synthesizes a massive pre-training dataset by converting existing, heterogeneous bounding-box and segmentation annotations into a unified image-caption data format, enabling robust contrastive language-image pre-training[69].

However, early RSVLMs primarily relied on short, rudimentary captions, which failed to capture the complex, multi-object semantic relationships inherent in wide-area satellite scenes. To address this, the LRSCLIP (and DGTRS-CLIP) framework utilizes a dual-granularity dataset known as LRS2M. By employing Multimodal Large Language Models (MLLMs)—strictly guided by carefully engineered prompt constraints to prevent geographic hallucination—the system generates highly detailed, long-text descriptions of satellite imagery. This dual-granularity approach enables complex zero-shot cross-modal retrieval, allowing users to query imagery using intricate geographic descriptions rather than basic keywords[71].

Alternatively, the GRAFT framework sidesteps the need for textual annotations entirely by utilizing internet-sourced, ground-level imagery as a semantic intermediary. By training a satellite image model to align its latent embeddings with the CLIP embeddings of co-located ground images, GRAFT transitively aligns natural text with overhead satellite imagery, enabling zero-shot open-vocabulary classification without requiring manually written captions during training[73]. Similarly, the GeoQuery system enables zero-shot natural-language retrieval by employing a two-stage semantic and visual search over a subset of "prompt-aligned proxy" tiles, circumventing the massive compute required for global contrastive training[45].

Spatial-Temporal Location Encoders

Concurrently, a specialized subset of models known as Location Encoders—such as SatCLIP, GeoCLIP, and CSP—address the critical need for continuous spatial representation. Traditional deep learning models struggle to assimilate geographic coordinates (latitude and longitude), often treating them as basic numerical inputs which exacerbates geographic distribution shifts[15].

Location encoders solve this by mapping raw geographic coordinates to high-dimensional, implicit neural embeddings. During pre-training, these models contrastively align coordinate representations with the visual features of co-located Sentinel-2 or natural imagery[15]. The resulting representations implicitly encode the physical, climatic, topographic, and anthropogenic characteristics of a given location on Earth. Incorporating these pre-trained location embeddings into downstream models drastically improves predictions for geographically dependent variables—such as dynamic PM2.5 air pollution estimation, species distribution modeling, and disease forecasting—even when raw satellite imagery is entirely absent at inference time, thereby enhancing cross-regional generalization[6].

Generative Foundation Models

Moving beyond discriminative and classification tasks, Generative Foundation Models for remote sensing utilize Denoising Diffusion Probabilistic Models (DDPMs) to synthesize highly accurate, physically realistic satellite imagery conditioned on various inputs.

DiffusionSat represents a major milestone in generative EO modeling. Trained on diverse, high-resolution datasets, it utilizes a novel 3D ControlNet architecture that strictly conditions the diffusion process on numerical geospatial metadata, including exact timestamps, geographic coordinates, cloud cover percentages, and Ground Sample Distance (GSD)[78]. This highly precise metadata conditioning enables DiffusionSat to solve complex generative inverse problems. For example, it performs multi-spectral super-resolution by upsampling coarse 10m-60m Sentinel-2 imagery to synthesize 0.3m high-resolution representations. Furthermore, it excels at temporal in-painting and predictive modeling; given pre-disaster imagery and specific metadata, DiffusionSat can realistically generate post-disaster floodscapes or predict urban development trajectories, providing invaluable synthetic data for disaster response training[79].

Similarly, the Text2Earth foundation model—built upon the 10-million sample Git-10M dataset—operates within a highly compressed Variational Autoencoder (VAE) feature space. This mathematical optimization enables the unbounded, global-scale generation of remote sensing scenes governed by strict spatial resolution guidance mechanisms, allowing users to iteratively construct massive synthetic landscapes via text prompts[82].

Segmentation and Prompt-Driven Analysis

The Segment Anything Model (SAM), pre-trained by Meta on over one billion natural image masks, exhibits profound zero-shot segmentation capabilities. However, when applied directly to remote sensing imagery in its "everything" mode, SAM frequently fails, producing erratic segmentations due to the dense clustering of small objects, extreme class imbalances, and atypical spectral signatures endemic to overhead imagery[13].

To bridge this operational gap without undertaking the computationally massive task of fully retraining the SAM architecture, researchers developed RSPrompter. Leveraging the paradigm of prompt learning, RSPrompter extracts intermediate feature maps from the frozen SAM image encoder and utilizes them to automatically generate semantically rich, category-aware prompts (including specific points, bounding boxes, and coarse masks) tailored for remote sensing instances. This autonomous prompt generation enables the foundational SAM framework to perform highly accurate, category-aware instance segmentation on satellite imagery without requiring manual human guidance[84].

For operational deployment across vast geographic extents, algorithmic wrapper pipelines such as Remote SAMsing manage the scaling of SAM2. By dividing massive orbital scenes into overlapping tiles, iteratively segmenting the terrain, and mathematically masking accepted regions to progressively simplify the scene, this approach successfully uncovers small, critical structures in extremely imbalanced datasets—such as identifying faint landslide scars that comprise less than 2% of total pixels in LULC datasets[87].

Challenges and Prospects

While Geospatial Foundation Models have indisputably advanced the frontiers of remote sensing, their transition from academic benchmarks to operational deployment at a planetary scale remains constrained by several fundamental challenges. Resolving these issues outlines the critical trajectory for future research.

Standardized Evaluation and the "State of the Art" Crisis

The rapid proliferation of GFMs has exposed a severe vulnerability in the remote sensing ecosystem: the pervasive absence of unified, rigorous evaluation protocols. Historically, new architectures were evaluated on a fragmented, inconsistent set of small, task-specific datasets (e.g., EuroSAT, RESISC45).

This lack of standardization has led to systemic ambiguity in the literature. In a sweeping, highly critical audit of 152 GFM papers published between 2019 and 2025, titled *"No One Knows the State of the Art in Geospatial Foundation Models,"* Corley et al. documented an alarming breakdown in scientific comparability[23]. The audit revealed 46 separate instances of severe cross-paper disagreements (reporting a variance of $\geq$ 10 percentage points) when evaluating the exact same model on the exact same benchmark using nominally identical protocols[23]. Furthermore, 94 out of the 126 papers with extractable data utilized entirely idiosyncratic, unreplicable pre-training data configurations, entirely confounding whether reported performance gains were the result of algorithmic innovation or simply exposure to disparate data[23]. Most critically, 39% of the surveyed papers failed to release their model weights, rendering downstream reuse and independent scientific verification impossible[23].

To resolve this crisis, the community is rapidly shifting toward comprehensive, standardized benchmark suites:

- **GEO-Bench:** Standardizes 12 diverse classification and segmentation tasks, heavily focusing on validating optical and multi-spectral representation learning[25].
- **PANGAEA:** A globally inclusive protocol spanning highly diverse domains (wildfires, marine environments, agriculture), sensors (Optical, SAR), and temporal settings. It provides a mathematically rigorous test for models claiming true multi-sensor superiority[38].
- **SustainFM:** Radically shifts the evaluation paradigm away from simple pixel accuracy metrics toward impact-driven deployment. It aligns 47 downstream tasks directly with the 17 UN Sustainable Development Goals (SDGs), emphasizing energy efficiency, out-of-distribution transferability, and carbon-efficient operational footprints as primary evaluation criteria[5].
- **Copernicus-Bench:** Provides 15 hierarchical downstream tasks specifically aligned with the Sentinel missions. Crucially, it integrates both traditional surface applications and

atmospheric tasks (e.g., cloud removal, precise air quality estimation), testing the true holistic capability of unified Earth system models[25].

Benchmark Suite	Primary Focus	Number of Tasks	Key Differentiator
GEO-Bench	Optical/Multispectral representation	12	Standardized pipeline for core LULC and segmentation.
PANGAEA	Multi-domain, global inclusivity	>10	Integrates extreme domain diversity (e.g., marine, wildfire).
SustainFM	UN Sustainable Development Goals	47	Evaluates energy efficiency, transferability, and real-world impact.
Copernicus-Bench	Sentinel mission integration	15	Hierarchical; unifies surface and atmospheric EO tasks.
REOBench	Distributional Robustness	6	Evaluates model degradation against 12 synthesized image corruptions.

Distributional Robustness and Domain Shift

High performance on independent and identically distributed (I.I.D) test sets frequently masks the profound fragility of GFMs when deployed in real-world, out-of-distribution (OOD) scenarios.
The REOBench (Robustness of Earth Observation Foundation Models) framework evaluates models against 12 synthesized, physically grounded image corruptions, including sensor noise, atmospheric haze, geometric rotations, and extreme scale shifts[94]. Empirical results from REOBench demonstrate that current GFMs—particularly those relying solely on Masked Image Modeling objectives—suffer severe performance degradation, with accuracy drops ranging from 1% to over 20% under environmental perturbations[94]. Conversely, Vision-Language

Models exhibit higher inherent robustness due to the broader semantic context ingested during their multimodal pre-training[94].
Similarly, the EarthShift benchmark systematically evaluates GFMs across geographic, temporal, and sensor shifts. Findings indicate that GFMs consistently suffer a 15-20% performance drop when evaluating out-of-distribution regions, regardless of their architectural size or specific pre-training strategy[97]. This underscores the reality that current SSL paradigms have not yet fundamentally solved the spatial generalization problem, necessitating a shift from optimizing pure accuracy to optimizing distributional robustness.

Modality Alignment and Heterogeneity

Fusing radically different physical modalities remains a mathematically non-trivial challenge. Combining the passive solar reflectance of optical sensors with the active phase and amplitude backscatter of SAR, or integrating continuous raster imagery with sparse vector geometries, requires complex architectural engineering[7]. Traditional "early fusion" techniques (e.g., simple band stacking) corrupt the distinct physical meaning of the inputs, while "late fusion" (decision ensembling) fails to capture deep, cross-modal correlations[36]. Future architectures must advance dynamic hypernetworks and implicit generative frameworks (such as TerraMind's TiM) that can mathematically align highly heterogeneous data streams—including atmospheric chemistry and topographic variables—within a unified, physics-aware latent space[61].

Computational Scalability and Energy Efficiency

The operational push toward high-resolution, global-scale analysis exacerbates the computational limits of current hardware. While Mamba architectures (SSMs) provide linear scaling[30], processing massive hyperspectral cubes and longitudinal temporal sequences remains highly carbon-intensive and computationally exclusionary[91]. Research must increasingly prioritize parameter-efficient fine-tuning (PEFT) methodologies. Techniques such as Low-Rank Adaptation (LoRA), hybrid parameter decomposition, and the deployment of explicitly distilled "tiny" foundation models are essential to allow the rapid adaptation of multi-billion parameter architectures on edge devices, or directly onboard orbital satellites under severe power constraints[3].

Conclusion

The integration of Foundation Models into Satellite Earth Observation marks a historic transition from fragmented, task-specific feature engineering to generalized, planetary-scale representation learning. By leveraging massive archives of unannotated geospatial data through advanced self-supervised paradigms—ranging from multi-scale Masked Image Modeling to temporal contrastive alignment—these models have demonstrated unprecedented adaptability across tasks encompassing precision agriculture, climate monitoring, and rapid disaster response.
Architectural innovations, notably the critical transition from quadratic-complexity Vision Transformers to linear-complexity State Space Models (Mamba), are unlocking the ability to process ultra-high-resolution, hyperspectral, and long-sequence temporal data at a global

scale. Concurrently, the emergence of multi-sensor unified models, vision-language spatial reasoners, and metadata-conditioned generative diffusion networks is vastly expanding the operational utility and accessibility of remote sensing data.
However, the trajectory of this field is critically threatened by an acute lack of rigorous, standardized evaluation, a documented vulnerability to real-world domain shifts, and the immense computational overhead required for deployment. The next era of geospatial artificial intelligence must pivot from an exclusive focus on raw predictive accuracy on curated datasets toward robust, mathematically sound, and impact-driven deployment. By embracing stringent benchmarking, transparent open-science practices, and parameter-efficient architectures, the remote sensing community can ensure that these powerful models remain accountable, accessible, and firmly aligned with global environmental and sustainability objectives.

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